# Aditya Vikram Singh

Software Engineer – Backend, Distributed Systems & AI Tooling
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## Summary

Software Development Engineer II at Autodesk, with 2+ years there (including a year-long internship) building distributed, real-time backend systems in C#/.NET, Python and TypeScript. Took a hackathon prototype to a funded production platform, cut end-to-end sync latency 10x, and built AI-driven test generation and distributed test infrastructure. Experienced with microservices, Kubernetes, gRPC and CI/CD, and working toward LLM applications: RAG, agents and evaluation.

## Experience

**Autodesk** — Pune, India
*Software Development Engineer II* — *Apr 2026 – Present*
*Software Development Engineer* — *Jul 2025 – Apr 2026*
*Software Engineering Intern* — *Jul 2024 – Jul 2025*

- Led a hackathon proof-of-concept for real-time collaboration across multiple modeling applications to a funded production platform, shipping private and public beta over three quarters.
- Designed a new data exchange format that cut end-to-end sync latency for large changes 10x (~15s to ~1.5s).
- Built a new in-memory BREP geometry representation that updates incrementally, enabling low-latency live editing.
- Designed and built an AI-driven test generation pipeline that derives test cases from product behavior, expanding coverage and reducing manual QA effort.
- Built a distributed end-to-end test lab in which a pool of workers pulls and runs cross-application tests, continuously validating the platform in CI.
- Decomposed a tightly coupled monolith into microservices (.NET, Kubernetes, gRPC, S3, Redis); in a team of 3, cut open issues by 75%.
- Parallelized the unit test suite and fixed static-state flakiness, cutting CI build-and-test time by 20 minutes; cut integration test runtime from 14 to 2 minutes.
- Extended a cloud data platform with 3D geometry (BREP and mesh) support, coordinating across 5 teams to ship it to early-access customers.
- Converted from a year-long internship to full-time, then promoted to SDE II within 9 months; presented the platform and led a class of 30 at Autodesk TechX conferences in Seattle and Las Vegas.

**Infosys Springboard** — Remote
*Machine Learning Intern* — *Apr 2024 – Jun 2024*

- Forecast energy consumption with ARIMA, LSTM and Prophet time-series models in a team of 5.
- Applied MLOps practices using MLflow, Apache Airflow, Docker and Kubernetes for model tracking and deployment.

**SCSIT, NIET** — Greater Noida, India
*Software Engineering Intern* — *May 2023 – Nov 2023*

- Built a document management platform in Java, Spring Boot and MySQL, with an admin interface and full CRUD, that automated manual workflows.

## Skills

**Languages:** Python, C#, TypeScript, JavaScript, SQL, Java
**AI / LLM:** LLM applications, RAG, vector databases, multi-agent systems, LangChain, AI-assisted development (Claude Code, GitHub Copilot)
**Machine Learning:** PyTorch, TensorFlow, time-series forecasting (LSTM, ARIMA, Prophet), MLflow, Airflow, MLOps
**Backend & Distributed Systems:** .NET, microservices, gRPC, REST APIs, Redis, real-time sync, data serialization formats, Flask, Spring Boot
**Cloud & DevOps:** Kubernetes, Docker, CI/CD, AWS S3, Azure, Google Cloud (GCP), test automation
**Domain:** 3D geometry (BREP, mesh), CAD/AEC interoperability

## Education

**Noida Institute of Engineering and Technology** — Greater Noida, India
*Integrated M.Tech, Computer Science and Engineering* — *Sep 2020 – Jun 2025*

## Certifications

Google Cloud Certified – Associate Cloud Engineer, Oct 2023 – Oct 2026 (Credly)
Microsoft Certified: Azure AI Fundamentals, 2023 (Credly)